

# High Frontier 4 All – Card List

Core, Modules 1 & 2

| Thrusters                       |               |      |          |                                                                                                                                                                             |                                                                                                                                                                            |                                 |
|---------------------------------|---------------|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Name                            | Spectral Type | Mass | Rad Hard | Thrust – Fuel (AB)                                                                                                                                                          | Requires                                                                                                                                                                   | Requires / Notes                |
| Ablative Plate                  |               | 1    | 7        | 2►2 (1)                                                                                                                                                                     |  /X                                                                                       |                                 |
| Ablative Nozzle                 | V             | 0    | 8        | 3►2 (1)                                                                                                                                                                     |  /X/   |                                 |
| De Laval Nozzle                 |               | 0    | 6        | 5•4 (2)                                                                                                                                                                     |  /     |                                 |
| Magnetic Nozzle                 | M             | 0    | 5        | 3•1 (3)                                                                                                                                                                     |                                                                                          |                                 |
| Dumbo                           |               | 0    | 7        | 6•5 (1)                                                                                                                                                                     |  /     |                                 |
| Timberwind                      | S             | 0    | 7        | 5•3 (1)                                                                                                                                                                     |  /X/   |                                 |
| Hall Effect                     |               | 2    | 5        | 3►2                                                                                                                                                                         |                                                                                          |                                 |
| Ion Drive                       | C             | 1    | 5        | 2►½ (1)                                                                                    |                                                                                          |                                 |
| Mass Driver                     |               | 3    | 8        | 4►3 Dirt                                                                                                                                                                    |                                                                                          |                                 |
| MPD T-wave                      | M             | 1    | 9        | 2►½ Dirt                                                                                                                                                                    |                                                                                          |                                 |
| Metastable Helium               |               | 1    | 0        | 5•1 (1)                                                                                                                                                                     | X                                                                                       |                                 |
| n- <sup>6</sup> Li Microfission | V             | 1    | 8        | 5•1/2 (1)                                                                                                                                                                   | X                                                                                     |                                 |
| Monatomic Plug Nozzle           |               | 0    | 6        | 4•3 (1)                                                                                                                                                                     |                                                                                        |                                 |
| Vortex Confined Nozzle          | M             | 0    | 7        | 4•2 (1)                                                                                                                                                                     |   |                                 |
| Photon Heliogyro                |               | 0    | 1        | 0►0   |                                                                                                                                                                            | Immune to Flare and Belt Rolls. |
| Electric Sail                   | C             | 0    | 2        | 2►0   |                                                                                                                                                                            | Immune to Flare and Belt Rolls. |
| Photon Kite Sail                |               | 0    | 0        | 0►0   |                                                                                                                                                                            | Immune to Flare and Belt Rolls. |
| Mag Sail                        | C             | 0    | 4        | 1►0   |                                                                                                                                                                            | Enter radiation: +1 flyby.      |
| Ponderomotive VASIMR            |               | 1    | 4        | 3►1 (1)                                                                                                                                                                     |                                                                                        |                                 |
| Pulsed Plasmoid                 | V             | 1    | 5        | 3►1  Dirt                                                                                |                                                                                                                                                                            |                                 |
| Pulsed Inductive                |               | 1    | 5        | 4►2                                                                                                                                                                         |                                                                                        |                                 |
| Dual-Stage 4-Grid               | C             | 0    | 5        | 2►1/3                                                                                    |                                                                                        |                                 |
| Re Solar Moth                   |               | 0    | 5        | 3►4 (2)                                                                                  |                                                                                                                                                                            |                                 |
| Colliding Beam H-B Fusion       | D             | 2    | 5        | 3•1/10 (2)                                                                               |   |                                 |

| Robonauts                       |            |      |          |                    |                             |                                                                                       |                                            |
|---------------------------------|------------|------|----------|--------------------|-----------------------------|---------------------------------------------------------------------------------------|--------------------------------------------|
| Name                            | Spec. Type | Mass | Rad Hard | Thrust – Fuel (AB) | ISRU                        | Requires                                                                              | Notes                                      |
| Blackbody-pumped Laser          |            | 0    | 5        |                    | Raygun 2                    |    |                                            |
| Fissile Aerosol Laser           | S          | 0    | 7        |                    | Raygun 2                    |    |                                            |
| Cat Fusion Z-pinch Torch        |            | 0    | 2        |                    | Buggy 2                     |    |                                            |
| H-B Cat Inertial                | D          | 2    | 2        | 4 • 1/3 (1)        | Missile 0                   |    |                                            |
| Flywheel Tractor                |            | 0    | 5        |                    | Buggy 3                     |    |                                            |
| Electrophoretic Sandworm        | D          | 1    | 5        |                    | Buggy 1                     |                                                                                       |                                            |
| Free Electron Laser             |            | 2    | 4        |                    | Raygun 1                    |    |                                            |
| Wakefield e-beam                | S          | 0    | 5        | 4 • 2 (2) Dirt     | Missile 1                   |    |                                            |
| Kuck Mosquito                   |            | 0    | 6        | 10 • 8 (2)         | Missile 3                   |    |                                            |
| Ablative Laser                  | D          | 0    | 6        | 3 ► 1 (3) ☀        | Missile 2                   |                                                                                       |                                            |
| MET Steamer                     |            | 1    | 4        | 5 ► 4 (1)          | Missile 2                   |    |                                            |
| Nanobot                         | C          | 0    | 5        |                    | Buggy 1                     |                                                                                       |                                            |
| Neutral Beam                    |            | 2    | 5        |                    | Raygun 2                    |  |                                            |
| D-D Fusion Inertial             | S          | 1    | 4        | 3 • ½ (1)          | Missile 1                   |  |                                            |
| Nuclear Drill                   |            | 1    | 6        | 5 ► 4 (1)          | Missile 3                   |  |                                            |
| Helical Railgun                 | D          | 2    | 8        | 5 ► 4 Dirt         | Raygun 0<br>OR<br>Missile 0 |  |                                            |
| Phase-Locked Diode Laser        |            | 1    | 3        |                    | Raygun 3                    |  |                                            |
| Lorentz-propelled Microprobe    | C          | 0    | 3        |                    | Raygun 3                    |                                                                                       | 1 reroll if fail 1 or more size rolls.     |
| Rock Splitter                   |            | 0    | 5        |                    | Buggy 4                     |  |                                            |
| MagBeam                         | V          | 1    | 5        | 4 ► 1 (1) ☀        | Raygun 3<br>OR<br>Missile 3 |                                                                                       | -1 ISRU & +3 Thrust if pushed by Powersat. |
| Solar-pumped MHD Exciplex Laser |            | 3    | 4        |                    | Raygun 2                    |  |                                            |
| Quantum Cascade Laser           | C          | 1    | 5        |                    | Raygun 0                    |  |                                            |
| Tungsten Resistojet             |            | 0    | 5        | 5 ► 4 (1)          | Missile 3                   |  |                                            |
| MITEE Arcjet                    | M          | 0    | 4        | 4 ► 2 (2)          | Missile 1                   |  |                                            |

| Refineries                      |               |      |          |          |                                                        |
|---------------------------------|---------------|------|----------|----------|--------------------------------------------------------|
| Name                            | Spectral Type | Mass | Rad Hard | Requires | Notes                                                  |
| Atomic Layer Deposition         |               | 2    | 2        | e        |                                                        |
| Ilmenite Semi-conductor film    | V             | 2    | 6        |          | Industrialize non-atmospheric size 8+: gain Powersat   |
| Basalt Fiber Spinning           |               | 3    | 5        | ☼/X/☾    |                                                        |
| Von Neumann Santa Claus Machine | S             | 2    | 5        |          | Colocated Crew / Robonaut / Colonist has ISRU -1       |
| Carbo-Chlorination              |               | 2    | 4        | e        |                                                        |
| Solar Carbotherm                | V             | 2    | 5        |          | No robonaut to industrialize Mercury/Venus/Earth zones |
| CVD Molding                     |               | 3    | 5        |          |                                                        |
| Carbonyl Volatilization         | M             | 2    | 5        |          | -3 to Colocated size rolls at S Sites.                 |
| Electroforming                  |               | 3    | 3        | e        |                                                        |
| Impact Mold Sinter              | M             | 3    | 5        |          | -1 to Colocated size rolls.                            |
| Fluidized-Bed                   |               | 3    | 4        | e        |                                                        |
| Atmospheric Scoop               | V             | 1    | 5        | e        | ☾; Adjacent/Colocated Aerostats are 💧                  |
| Foamglass Sintering             |               | 3    | 4        | e        |                                                        |
| Laser-heated Pedestal Growth    | D             | 1    | 5        | e        | -1 to Colocated Raygun size rolls.                     |
| Froth Flotation                 |               | 3    | 5        | e        |                                                        |
| Femtochemistry                  | D             | 2    | 8        |          | Doubles Colocated Site Refuel FTs.                     |
| In-Situ Leaching                |               | 3    | 3        | ☼/X/☾    |                                                        |
| Termite Nest                    | D             | 1    | 3        | ☼/X/☾    | Claim Size 2+ busted Site as Op                        |
| ISRU Sabatier                   |               | 3    | 4        | e        |                                                        |
| Biophytolytic Algal Farm        | V             | 3    | 5        |          | -2 to Colocated size rolls at D Sites.                 |
| Magma Electrolysis              |               | 3    | 3        | e        |                                                        |
| Ionosphere Lasing               | V             | 2    | 6        |          | Industrialize atmospheric site: gain Powersat          |
| Supercritical Drying            |               | 1    | 5        | X        |                                                        |
| Solid Flame                     | H             | 0    | 5        | X        | Colocated Industrialization is a Free Action.          |

| Crew                |                            |       |      |          |              |                       |                                                                                       |
|---------------------|----------------------------|-------|------|----------|--------------|-----------------------|---------------------------------------------------------------------------------------|
| Faction             | Name                       | Clout | Mass | Rad Hard | Thrust (AB)  | ISRU                  | Faction Abilities                                                                     |
| Green: Equality     | ESA Space Unionists        | F     | 1    | 4        | 12 ► 8 (2)   | Missile 4             | Powersat (+1 Push to any craft).                                                      |
|                     | Anonymous P2P              | E     | 1    | 4        | 14 ► 8 (2)   | Buggy 4<br>Missile 4  | FINAO costs 3 Aqua.                                                                   |
| White: Honor        | NASA Astronauts            | D     | 1    | 4        | 14 • 8 (2)   | Missile 4             | 1 Aqua for each boost                                                                 |
|                     | ISRO Glavcosmonauts        | G     | 1    | 4        | 11 • 14 (2)  | Raygun 4<br>Missile 4 | Double Site Refuel when carrying Glory chit.                                          |
| Purple: Authority   | Taikonauts                 | C     | 1    | 4        | 14 • 9 (2)   | Missile 4             | Felonious                                                                             |
|                     | Roscosmos                  | B     | 1    | 5        | 15 • 10 (2)  | Raygun 4<br>Missile 4 | 1 Aqua for each Claim / Industrialize                                                 |
| Red: Freedom        | Shimizu Corp Entrepreneurs | M     | 1    | 3        |              | Raygun 4<br>Buggy 4   | No hand limit for auctions                                                            |
|                     | NASRDA Astronauts          | L     | 1    | 4        | 7 • 11 Dirt  | Missile 4             | Dirt Refuel 1 FT (7 for non-crew thrusters) at LEO / Home Orbit as Free Action 1/turn |
| Orange: Unity       | United Nations Cosmonauts  | A     | 1    | 4        | 12 • 9 (2)   | Buggy 4<br>Missile 4  | Start game with +2 Aqua.                                                              |
|                     | B612 Foundation            | H     | 1    | 3        | 15 • 10 (2)  | Missile 4<br>Raygun 4 | 1 re-roll per Prospect Op when using Raygun.                                          |
| Grey: Individuality | SpaceX                     | J     | 1    | 4        | 15 • 10 (2)  | Missile 4             | Always win Auction ties.                                                              |
|                     | Norse Astronauts           | K     | 1    | 3        | 15 • 14 Dirt | Buggy 4<br>Missile 4  | Repair any Glitch anywhere.                                                           |

| Generators                                |               |                                                                                                                                                                         |      |          |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                |
|-------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                      | Spectral Type | Subtype                                                                                                                                                                 | Mass | Rad Hard | Requires                                                                                                                                                                                                                                                                                                                                                                                                                                     | Modifiers                                                                                                                                                                                      |
| AMTEC Thermoelectric                      |               |       | 1    | 6        |                                                                                                                                                                                                                                                                        |                                                                                                             |
| JTEC H <sub>2</sub> Thermoelectric        | C             |       | 1    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |  ,                       |
| Brayton Turbine                           |               |       | 1    | 5        |  /                                                                                                 |                                                                                                             |
| O'Meara LSP Paralens                      | C             |                                                                                        | 0    | 10       |                                                                                                                                                                                                                                                                                                                                                                                                                                              |  ,  , ½ Fuel Consumption |
| Cascade Photovoltaic                      |               |                                                                                        | 3    | 1        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
| Buckyball C <sub>60</sub> Photovoltaic    | C             |                                                                                        | 2    | 3        |                                                                                                                                                                                                                                                                                                                                                                                                                                              | +1 Thrust,                                                                                                  |
| Cascade Thermoacoustic                    |               |                                                                                        | 0    | 5        |                                                                                                                                                                                      |                                                                                                             |
| Dusty Plasma MHD                          | V             |                                                                                        | 1    | 6        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
| Catalyzed Fission Scintillator            |               |       | 0    | 7        |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                |
| Diamonoid Electro-dynamic Tether          | D             |       | 2    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                |
| Ericsson Engine                           |               |                                                                                        | 1    | 5        |  /                                                                                                                                                                                   |                                                                                                             |
| Nanocomposite Thermoelectric              | C             |                                                                                        | 1    | 5        |  /  /                                                                                                                                                                                 | +1 Thrust,                                                                                                  |
| Flywheel Compulsator                      |               |       | 4    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
| Superconducting Adductor                  | M             |       | 1    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                |
| H <sub>2</sub> – O <sub>2</sub> Fuel Cell |               |                                                                                        | 4    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
| Microbial Fuel Cell                       | C             |       | 2    | 5        |                                                                                                                                                                                                                                                                                                                                                                                                                                              | +1 Thrust,                                                                                                  |
| In-core Thermionic                        |               |       | 0    | 6        |  /                                                                                                 |                                                                                                             |
| Z-pinch Microfission                      | S             |       | 2    | 9        |                                                                                                                                                                                                                                                                                                                                                           | +4 Thrust,                                                                                                  |
| Magnetoshell Plasma Parachute             |               |                                                                                       | 0    | 5        |                                                                                                                                                                                                                                                                                                                                                                                                                                              | -2 Thrust,  , safe aerobrace. Cannot support Bernals or during Industrialization.                          |
| Granular Rainbow Corral                   | S             |                                                                                      | 0    | 10       |                                                                                                                                                                                                                                                                                                                                                                                                                                              |  , safe aerobrace.                                                                                        |
| Marx Capacitor Bank                       |               |                                                                                      | 1    | 4        |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                |
| Casimir Battery                           | M             |                                                                                      | 0    | 3        |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                |
| Optoelectric Nuclear Battery              |               |                                                                                      | 2    | 6        |                                                                                                                                                                                                                                                                    |                                                                                                           |
| Nuclear-Pumped Excimer Flashlamp          | V             |                                                                                      | 0    | 8        |  /     | +1 Thrust,                                                                                                |
| Photon Tether Rectenna                    |               |                                                                                      | 2    | 2        |                                                                                                                                                                                                                                                                                                                                                                                                                                              | -1 Thrust,                                                                                                |
| Palmer LSP Aerosol Lens                   | V             |                                                                                      | 0    | 10       |                                                                                                                                                                                                                                                                                                                                                                                                                                              | +1 Thrust,  ,        |
| Radioisotope Stirling                     |               |                                                                                      | 3    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              | -2 Thrust                                                                                                                                                                                      |
| Triggered Decay Nuclear Battery           | M             |   | 1    | 6        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                |
| Rankine MHD                               |               |                                                                                      | 0    | 4        |  /                                                                                         |                                                                                                           |
| MHD Open-Cycle                            | M             |   | 0    | 4        |                                                                                                                                                                                                                                                                                                                                                         | +1 Thrust,                                                                                                |
| Rankine Solar Dynamic                     |               |                                                                                      | 2    | 4        |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
| Rankine Multiphase                        | D             |   | 0    | 4        |  /  /                                                                                                                                                                           |                                                                                                                                                                                                |
| Solar Stirling                            |               |                                                                                      | 3    | 5        |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
| Thermo-Photovoltaic                       | C             |                                                                                      | 2    | 4        |                                                                                                                                                                                                                                                                                                                                                                                                                                              |  , +1 Thrust                                                                                              |

## Radiators

| Name                             | Spectral Type | Light Side                                                                          |      |          | Heavy Side                                                                           |      |          | Notes                                                                                                                           |
|----------------------------------|---------------|-------------------------------------------------------------------------------------|------|----------|--------------------------------------------------------------------------------------|------|----------|---------------------------------------------------------------------------------------------------------------------------------|
|                                  |               | Therms                                                                              | Mass | Rad Hard | Therms                                                                               | Mass | Rad Hard |                                                                                                                                 |
| Bubble Membrane                  |               |    | 0    | 1        |    | 1    | 0        |                                                                                                                                 |
| Electrostatic Membrane           | C             |    | 0    | 3        |    | 1    | 2        |                                                                                                                                 |
| Dielectric X-Ray Window          |               |    | 4    | 7        |    | 5    | 7        |                                                                                                                                 |
| Graphene Crystal X-Ray Mirror    | D             |    | 0    | 7        |    | 1    | 7        |                                                                                                                                 |
| ETHER Charged Dust               |               |    | 1    | 1        |    | 2    | 1        |                                                                                                                                 |
| Curie Point                      | M             |    | 0    | 5        |    | 1    | 3        |                                                                                                                                 |
| Li Heatsink Fountain             |               |    | 2    | 6        |    | 3    | 6        | Heavy side 1 use only                                                                                                           |
| Thermochemical Heatsink Fountain | S             |    | 1    | 6        |    | 2    | 6        | Heavy side 1 use only                                                                                                           |
| Magnetocaloric Refrigerator      |               |    | 2    | 6        |    | 3    | 6        | Requires  generator<br>Can cool own supports |
| Nuclear Fuel Spin Polarizer      | S             |    | 0    | 5        |    | 1    | 4        |                                                                                                                                 |
| Microtube Array                  |               |    | 2    | 1        |    | 3    | 0        |                                                                                                                                 |
| Marangoni Flow                   | C             |  | 0    | 2        |  | 1    | 1        |                                                                                                                                 |
| Mo / Li Heat Pipe                |               |  | 1    | 2        |  | 2    | 1        |                                                                                                                                 |
| Tin Droplet                      | M             |  | 0    | 3        |  | 1    | 2        |                                                                                                                                 |
| Qu Tube                          |               |  | 1    | 4        |  | 4    | 2        |                                                                                                                                 |
| ANDR / In Dream Pipe             | M             |  | 0    | 3        |  | 1    | 2        |                                                                                                                                 |
| Salt-cooled Reflux Tube          |               |  | 3    | 4        |  | 4    | 4        |                                                                                                                                 |
| Buckytube Filament               | C             |  | 0    | 4        |  | 1    | 3        |                                                                                                                                 |
| SS / NaK Pumped Loop             |               |  | 2    | 1        |  | 4    | 0        |                                                                                                                                 |
| Hula-Hoop                        | M             |  | 1    | 6        |  | 2    | 6        |                                                                                                                                 |
| Steel / Pb – Bi Pumped Loop      |               |  | 2    | 3        |  | 4    | 2        |                                                                                                                                 |
| Pulsating Heat Pipe              | M             |  | 1    | 1        |  | 2    | 1        |                                                                                                                                 |
| Ti / K Heat Pipe                 |               |  | 2    | 3        |  | 4    | 2        |                                                                                                                                 |
| Flux-pinned Superthermal         | M             |  | 1    | 5        |  | 2    | 5        |                                                                                                                                 |

| Reactors                              |               |                                                                                                                                                                                                                                                       |      |          |                                                                                                                                                                           |                                                                  |
|---------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Name                                  | Spectral Type | Type                                                                                                                                                                                                                                                  | Mass | Rad Hard | Requires                                                                                                                                                                  | Notes                                                            |
| Cermet NERVA Fission                  |               |                                                                                                                                                                      | 1    | 7        |                                                                                                                                                                           | +3 Thrust                                                        |
| Pulsed NTR Fission                    | S             |                                                                                                                                                                      | 0    | 7        |                                                                                         | ¼ Fuel Consumption                                               |
| D-D Fusion Magneto-Inertial           |               |                                                                                                                                                                      | 1    | 3        |                                                                                                                                                                           | +2 Thrust, ☼                                                     |
| H-B Fusion Reciprocating Plasmoid     | D             |                                                                                                                                                                      | 0    | 7        |                                                                                         | +4 Thrust<br>½ Fuel Consumption                                  |
| D-T Fusion Tokamak                    |               |                                                                                                                                                                      | 3    | 5        |                                                                                         | +3 Thrust<br>½ Fuel Consumption                                  |
| Antimatter GDM                        | H             |    | 0    | 9        |                                                                                         | ¼ Fuel Consumption                                               |
| D-T Gun Fusion                        |               |                                                                                                                                                                      | 1    | 9        |                                                                                         | +1 Thrust<br>¼ Fuel Consumption                                  |
| Macron Blowpipe Fusion                | H             |    | 0    | 10       |                                                                                         | 1/10 Fuel Consumption                                            |
| Lyman Alpha Trap                      |               |                                                                                     | 2    | 4        |                                                                                                                                                                           | -2 Thrust                                                        |
| Free Radical Hydrogen Trap            | S             |                                                                                     | 0    | 4        |                                                                                         | ½ Fuel Consumption                                               |
| Metallic Hydrogen                     |               |                                                                                                                                                                      | 0    | 2        |                                                                                         | +4 Thrust                                                        |
| Fission-augmented D-T Inertial Fusion | V             |                                                                                                                                                                     | 1    | 7        |                                                                                        | +7 Thrust                                                        |
| Mini-Mag RF Paul Trap                 |               |                                                                                                                                                                    | 1    | 2        |                                                                                                                                                                           | -3 Thrust, ☾<br>-2 ISRU for Colocated ISRU at Aerostats          |
| Ultracold Neutrons                    | C             |                                                                                 | 1    | 5        |                                                                                                                                                                           | ☾<br>-2 ISRU for Colocated ISRU at Aerostats                     |
| Pebble Bed Fission                    |               |                                                                                                                                                                    | 1    | 6        |                                                                                       | +4 Thrust                                                        |
| VCR Light Bulb Fission                | S             |                                                                                                                                                                    | 0    | 6        |                                                                                       | +2 Thrust<br>½ Fuel Consumption                                  |
| Penning Trap                          |               |                                                                                 | 2    | 3        |                                                                                                                                                                           |                                                                  |
| <sup>3</sup> He-D Fusion Mirror Cell  | H             |                                                                                                                                                                    | 1    | 5        |                                                                                       | ¼ Fuel Consumption                                               |
| Project Orion                         |               |                                                                                                                                                                    | 6    | 8        |                                                                                       | +7 Thrust                                                        |
| Project Valkyrie                      | S             |                                                                                                                                                                    | 1    | 9        |   | +8 Thrust<br>¼ Fuel Consumption<br>Decom. Colocated cards RH < 4 |
| Rubbia Thin Film Fission Hohlraum     |               |                                                                                                                                                                    | 1    | 5        |                                                                                       | ½ Fuel Consumption                                               |
| Positronium Bottle                    | M             |                                                                                 | 0    | 9        |                                                                                       | ½ Fuel Consumption                                               |
| Supercritical Water Fission           |               |                                                                                                                                                                    | 1    | 7        |                                                                                       | +3 Thrust                                                        |
| H- <sup>6</sup> Li Fusor              | V             |                                                                                 | 0    | 4        |                                                                                                                                                                           | +2 Thrust                                                        |

| Bernals                        |      |          |             |       |                                                                                                                                                                     |                                                                                      |                                                                                                                                            |
|--------------------------------|------|----------|-------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Name                           | Mass | Rad Hard | Thrust (AB) | Orbit | Supports                                                                                                                                                            | Promotes At                                                                          | Ability                                                                                                                                    |
| L4 Antimatter Factory          | 10   | 9        | 3►3 Dirt    | L4    |   |    | Home: Your Crew has on-board  .                         |
| Antimatter Lab                 | 10   | 10       | 3►3 Dirt    |       |   |                                                                                      | Your Crew has an on-board Reactor (any).                                                                                                   |
| L5s Cancer Hospital            | 10   | 6        | 3►3 Dirt    | L5s   |                                                                                    |    | Home: Immune to Budget Cuts.                                                                                                               |
| Cancer Lab                     | 12   | 7        | 3►3 Dirt    |       |                                                                                    |                                                                                      | +1 VP for each Colony dome. Your Crew and Human Colonists have rad-hard of at least 7.                                                     |
| L1 Climate Control Bernal      | 10   | 8        | 3►3 Dirt    | L1    |                                                                                    |    | Home: You are always 1 <sup>st</sup> player.                                                                                               |
| Climate Control Lab            | 12   | 8        | 3►3 Dirt    |       |                                                                                    |                                                                                      | +2 VP per Dirtsides for this Bernal.                                                                                                       |
| L2 Collimator Bernal           | 10   | 8        | 3►3 Dirt    | L2    |                                                                                    |    | Home: Gain Powersat.                                                                                                                       |
| Collimator Lab                 | 10   | 9        | 3►3 Dirt    |       |                                                                                    |                                                                                      | Gain Powersat. Your Powersat push includes a Bonus Pivot.                                                                                  |
| SSO Diplomatic                 | 10   | 6        | 3►3 Dirt    | SSO   |                                                                                    |    | Home: Your delegates in your color Ideology are +1 VP each.                                                                                |
| Diplomatic Lab                 | 10   | 7        | 3►3 Dirt    |       |                                                                                    |                                                                                      | Your delegates in the assembly are +1 VP each.                                                                                             |
| GEO Elevator Bernal            | 10   | 6        | 3►3 Dirt    | GEO   |                                                                                   |   | Home: Boost direct to Bernal without doubling Boost cost.                                                                                  |
| Space Elevator Lab             | 10   | 7        | 3►3 Dirt    |       |                                                                                  |                                                                                      | Your factory-assist landings/liftoffs treat Lander Burns as normal spaces.                                                                 |
| L3 Lofstrom Loop Microgravity  | 10   | 8        | 3►3 Dirt    | L3    |                                                                                  |  | Home: Boost direct to Bernal without doubling boost cost.                                                                                  |
| Lofstrom Loop Microgravity Lab | 10   | 9        | 3►3 Dirt    |       |                                                                                  |                                                                                      | Your factory-assist landings/liftoffs treat Lander Burns as normal spaces.                                                                 |
| L4s Pharmaceuticals Bernal     | 10   | 6        | 3►3 Dirt    | L4s   |                                                                                  |  | Home: Gain Skunkworks.                                                                                                                     |
| Pharmaceuticals Lab            | 10   | 7        | 3►3 Dirt    |       |                                                                                  |                                                                                      | Gain Skunkworks & impose hand limit on all opponents.                                                                                      |
| L5 Solar Cell Factory          | 10   | 6        | 3►3 Dirt    | L5    |                                                                                  |  | Home: +1 to Net Thrust of your Spacecraft that use  . |
| Solar Cell Lab                 | 10   | 8        | 3►3 Dirt    |       |                                                                                  |                                                                                      | +2 to Net Thrust of your Spacecraft that use  .       |
| Tourism Cyler                  | 10   | 7        | 3►3 Dirt    | Cyler |                                                                                  |  | Home: Can allow any Spacecraft to forego Belt Rolls around Earth.                                                                          |
| Tourism Hotel                  | 10   | 7        | 3►3 Dirt    |       |                                                                                  |                                                                                      | +2 VP per Dirtsides for this Bernal.                                                                                                       |

**GW / TW Thrusters**

| Name                                        | Type / Promote                                                                      | Thrust – Fuel (AB) | Mass | Rad Hard | Requires                                                                                                                                                                                                                                                                                                                                                                                                                                              | Notes                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------|--------------------|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Amat-Catalyzed Fission-Fusion               | S                                                                                   | 10 • 1/3 (+2)      | 3    | 9        |                                                                                                                                                                                                                                                                                  |                                         |
| Amat-Initiated H-B Magnetic-Inertial        |    | 8 • 0 (+5)         | 1    | 10       |                                                                                                                                                                                                                                                                                  | ★ Mini-Black Hole Future                |
| Dense Plasma H-B Focus Fusion               | D                                                                                   | 2 • 1/10 (+5)      | 1    | 8        |                                                                                                                                                                                                                                                                                                                                                                     |                                         |
| Crossfire H-B Focus Fusion                  |    | 3 • 0 (+9)         | 1    | 9        |                                                                                                                                                                                                                                                                                                                                                                     | ★ Protium Fusion Future                 |
| Levitated Dipole <sup>6</sup> Li-H Fusion   | V                                                                                   | 2 • 0 (+4)         | 4    | 5        |                                                                                                                                                                                                                                                                                                                                                                    |                                         |
| Dusty Plasma                                |    | 2 • 0 (+9)         | 2    | 8        |                                                                                                                                                                                               | ★ Mass Beam Future                      |
| Mini-Mag Orion Z-pinch Fusion               | M                                                                                   | 12 • 1/3           | 1    | 9        |                                                                                                                                                                                                                                                                                  |                                         |
| Solem Medusa Tugged Orion                   |    | 9 • 0 (+3)         | 6    | 9        |                                                                                                                                                                                                                                                                                                                                                                     | ★ Lithiated Ammonia Ice Starship Future |
| Salt-Water Zubrin                           | S                                                                                   | 14 • ½ (+1)        | 1    | 6        |                                                                                                             |                                         |
| Zubrin-GDM                                  |    | 6 • 0 (+8)         | 3    | 8        |                                                                                                              | ★ Spacefaring Future                    |
| Spheromak <sup>3</sup> He-D Magnetic Fusion | H                                                                                   | 6 • 1/10 (+6)      | 4    | 8        |                                                                                                              |                                         |
| Colliding FRC <sup>3</sup> He-D Fusion      |  | 4 • 0 (+10)        | 3    | 9        |  <br>2x:<br> /  /  | ★ Enzmann Starship Future               |
| VISTA D-T Inertial Fusion                   | H                                                                                   | 9 • ¼ (+3)         | 5    | 8        |                                                                                                                                                                                                                                                                              |                                         |
| Daedelus <sup>3</sup> He-D Inertial Fusion  |  | 11 • 0 (+8)        | 5    | 9        |                                                                                                                                                                                         | ★ Fusion Candle Future                  |

## Freighters

| Name                                    | Type / Promote                                                                      | Thrust - Fuel                                                                                                                                                                                                                                                                                                                                             | Mass | Rad Hard | Load Limit | Notes                                                                                                                                                                                                                        |
|-----------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fission-heated Steam                    | Any                                                                                 | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 0    | 5        | 2          | Factory Loading only.<br>                                                                                                                 |
| Fission GCR                             |    | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 1    | 6        | 6          | Can liftoff sites w/Size < 6 without factory-assist.<br><br>★ Exoplanet Hunt Future                                                       |
| Fusion Fragment Sail                    | V                                                                                   | 1 ▶ 0                                                                                                                                                                                                                                                                    | 2    | 1        | 2          | Immune to flares & radiation belts.                                                                                                                                                                                          |
| Antiproton Sail and Harvester           |    | 1 ▶ 0<br><br><br><br> | 2    | 9        | 6          | +1 net thrust if start on radiation belt.<br>★ Antimatter Future                                                                                                                                                             |
| HIIPER Beam Rider                       | V                                                                                   | 1 ▶ 0                                                                                                                                                                                                                                                                    | 4    | 4        | 4          |                                                                                                                                           |
| Magnetic Mirror Beam Rider              |    | 1 ▶ 0<br><br><br>                                                                                      | 3    | 6        | 5          | Can liftoff sites w/Size < 6 without factory-assist.<br><br>★ Star Wisp Future                                                            |
| Inflatable Solar-heated                 | Any                                                                                 | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 0    | 5        | 2          | Factory Loading only.<br>Requires Push Factory past Ceres zone.                                                                                                                                                              |
| Archimedes Palmer Lens                  |    | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 0    | 10       | 5          | Factory Loading only.<br>Requires Push Factory past Jupiter zone.<br>★ Terraform Future                                                                                                                                      |
| Poodle Steam                            | Any                                                                                 | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 0    | 7        | 2          | Factory Loading only.<br>+2 net thrust if starts from factory.                                                                                                                                                               |
| D-nanotube Dirt Launcher                |  | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 0    | 9        | 3          | ★ Beanstalk Future                                                                                                                                                                                                           |
| Rotary Dirt Launcher                    | Any                                                                                 | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 5    | 4        | 2          |  <br>Factory Loading only.                         |
| KESTS Hoop Dirt Launcher                |  | 1 ▶ 0                                                                                                                                                                                                                                                                                                                                                     | 15   | 8        | 15         |  <br>Factory Loading only.<br>★ Beehive Ark Future |
| Z-pinch D-T / <sup>6</sup> Li Fusion    | V                                                                                   | 1 ▶ 0                                                                                                                                                                                                                                                                  | 5    | 3        | 3          |                                                    |
| Z-pinch <sup>3</sup> He-D Target Fusion |  | 1 ▶ 0 <br>                                                                                                                                                                          | 2    | 6        | 6          |  <br>★ Golden Apples Future                        |

## Colonists (Page 1 of 2)

| Name                         | Type / Promote                                                                      | Thrust – Fuel (AB)                                                                                                                                                               | Mass | Rad Hard | Delegate                                                                            | ISRU               | Notes                                                                                                                                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|-------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Babbage Halbonauts           | H Robot                                                                             |                                                                                                                                                                                  | 2    | 5        |                                                                                     | Buggy 4            | <b>Engineer</b> (ET Produce 2 Cards)                                                                                                                                                                                                                                  |
| Utility Fog Halbonaut        |    |                                                                                                                                                                                  | 2    | 5        |                                                                                     | Buggy 2            | <b>Engineer</b> (ET Produce 2 Cards)<br>All of your stacks are Glitch-free.<br>★ Uplift Future                                                                                                                                                                        |
| Biomechs                     | Human                                                                               |                                                                                                                                                                                  | 2    | 4        |    | Buggy 3            | <b>Miner</b> (1 free Site Refuel / Turn)                                                                                                                                                                                                                              |
| Group Mind Immortalists      |    |                                                                                                                                                                                  | 2    | 5        |    | Buggy 2            | <b>Miner</b> (1 free Site Refuel / Turn)<br>Gain faction privileges on both sides of your Crew card.<br>★ Pan Sapiens Future                                                                                                                                          |
| Botany Bay Convicts          | Human                                                                               |                                                                                                                                                                                  | 2    | 4        |    | Buggy 4            | <b>Miner</b> (1 free Site Refuel / Turn)                                                                                                                                                                                                                              |
| Soldier Caste                |    |                                                                                                                                                                                  | 2    | 9        |    | Buggy / Raygun 2   | <b>Miner</b> (1 free Site Refuel / Turn)<br> All your Humans can commit Felonies even if other Humans present.<br>★ Secession Future                                               |
| Boyle Engineering Collective | Human                                                                               |                                                                                                                                                                                  | 3    | 5        |    | Buggy 3            | <b>Prospector</b> (Prospect/Promotion)                                                                                                                                                                                                                                |
| Martian Assembly             |    |                                                                                                                                                                                  | 3    | 6        |    | Buggy 2            | <b>Prospector</b> (Prospect/Promotion)<br>Acts as a Freighter to build Space Elevators.<br>★ Beanstalk Future                                                                                                                                                         |
| Calypso 2 Seed Sail          | Human                                                                               | 0 ▶ 0<br>  | 1    | 3        |  | Raygun / Missile 4 | <b>Prospector</b> (Prospect/Promotion)<br>Can't enter aerobrake.                                                                                                                                                                                                      |
| Wet-Nano Seed Sail           |  | 1 ▶ 0<br>  | 1    | 5        |  | Raygun / Missile 1 | <b>Prospector</b> (Free Prospect/Promotion)<br>-2 to Colocated size rolls on Synodic Comets.<br>Can't enter aerobrake.<br>★ New Venus Future                                                                                                                          |
| Heavy Water Survivalists     | Human                                                                               |                                                                                                                                                                                  | 2    | 5        |  | Raygun 4           | <b>Engineer</b> (ET Produce 2 Cards)                                                                                                                                                                                                                                  |
| New Attica Secessionists     |  |                                                                                                                                                                                  | 2    | 6        |  | Raygun 2           | <b>Engineer</b> (ET Produce 2 Cards)<br>Boost costs doubled for all opponents.<br>★ Secession Future                                                                                                                                                                  |
| House of Saud                | Human                                                                               |                                                                                                                                                                                  | 2    | 3        |  | Buggy 3            | <b>Miner</b> (1 free Site Refuel / Turn)                                                                                                                                                                                                                              |
| Iceworms                     |  |                                                                                                                                                                                  | 2    | 4        |  | Buggy 1            | <b>Miner</b> (1 free Site Refuel / Turn)<br>Performs Epic Hazards as Free Actions – not Decommissioned on failure.<br>★ Submariner Future                                                                                                                             |
| Juiced Cosmonauts            | Human                                                                               | 10 • 4<br>(1)                                                                                                                                                                    | 1    | 4        |  | Missile 3          | <b>Prospector</b> (Prospect/Promotion)                                                                                                                                                                                                                                |
| Rental Body Guild            |  | 12 • 4<br>(1)                                                                                                                                                                    | 1    | 6        |  | Missile 2          | <b>Prospector</b> (Prospect/Promotion)<br> -1 to Colocated size rolls. <br>★ ET Life Future |
| Lloyd's Salvage Co.          | Human                                                                               |                                                                                                                                                                                  | 1    | 5        |  |                    | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)                                                                                                                                                                                                                 |
| Svalbard Caretakers          |  |                                                                                                                                                                                  | 1    | 6        |  |                    | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)<br>-1 on all size rolls on Synodic Sites.<br>★ Dyson Bubble Future                                                                                                                                              |

## Colonists (Page 2 of 2)

| Name                      | Type / Promote                                                                      | Thrust – Fuel (AB) | Mass | Rad Hard | Delegate                                                                            | ISRU      | Notes                                                                                                                                                                                              |
|---------------------------|-------------------------------------------------------------------------------------|--------------------|------|----------|-------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Malcolm                   | Human                                                                               |                    | 1    | 3        |    |           | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)                                                                                                                                              |
| Renaissance Man           |    |                    | 1    | 4        |    |           | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)<br>Research Auction: Search through one deck and choose card to auction.<br>★ Artificial Consciousness Future                                |
| Microgravity Pantrophists | Human                                                                               |                    | 3    | 5        |    |           | <b>Engineer</b> (ET Produce 2 Cards)                                                                                                                                                               |
| Blue Goo Sybonts          |    |                    | 3    | 6        |    |           | <b>Engineer</b> (ET Produce 2 Cards)<br>ET Produce C cards at any Factory.<br>★ SETI Future                                                                                                        |
| Programmable Matter       | D Robot                                                                             | 2 ▶ 4 (1)          | 1    | 4        |                                                                                     | Missile 3 | <b>Prospector</b> (Prospect/Promotion)<br>                                                                      |
| Neumann Matter            |    | 4 ▶ 4 (1)          | 1    | 5        |                                                                                     | Missile 1 | <b>Prospector</b> (Prospect/Promotion)<br> All of your stacks are Glitch-free.<br>★ Uplift Future               |
| Rock Rats Miners' Union   | Human                                                                               |                    | 3    | 5        |    | Raygun 3  | <b>Miner</b> (1 free Site Refuel / Turn)<br>                                                                    |
| Alchemist Aviatrices      |    |                    | 3    | 6        |    | Raygun 0  | <b>Miner</b> (1 free Site Refuel / Turn)<br> Factory Refuel: 2x isotope fuel.<br>★ Aerostat Future              |
| Security System           | D Robot                                                                             |                    | 1    | 4        |                                                                                     |           | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)                                                                                                                                              |
| Frankenstein Navigator    |  |                    | 1    | 5        |                                                                                     |           | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)<br> FINAO costs halved (round down).<br>★ Uplift Future |
| Siren Cybernautics Inc.   | Human                                                                               |                    | 3    | 5        |  |           | <b>Engineer</b> (ET Produce 2 Cards)                                                                                                                                                               |
| Josephson Implants        |  |                    | 3    | 6        |  |           | <b>Engineer</b> (ET Produce 2 Cards)<br>FINAO costs halved (round down).<br>★ Supreme Cult Future                                                                                                  |
| Smart Pets                | D Robot                                                                             |                    | 0    | 3        |                                                                                     | Buggy 3   | <b>Miner</b> (1 free Site Refuel / Turn)                                                                                                                                                           |
| Creeper Neogen            |  |                    | 0    | 6        |                                                                                     | Buggy 2   | <b>Miner</b> (1 free Site Refuel / Turn)<br>All of your stacks are Glitch-free.<br>★ Uplift Future                                                                                                 |
| Transorbital Railworkers  | Human                                                                               |                    | 2    | 4        |  |           | <b>Engineer</b> (ET Produce 2 Cards)                                                                                                                                                               |
| Kaluga Naniteers          |  |                    | 2    | 5        |  |           | <b>Engineer</b> (ET Produce 2 Cards)<br>Aqua from a Free Market is doubled.<br>★ TNO Future                                                                                                        |
| Vatican Observers         | Human                                                                               |                    | 1    | 4        |  |           | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)                                                                                                                                              |
| Eugenic Pilgrims          |  |                    | 1    | 5        |  |           | <b>Industrialist</b> (Indust. / Anchor / Nanofacture)<br>Your faction privilege not lost during Anarchy.<br>-1 to Colocated size rolls on Synodic Comets.<br>★ Footfall Future                     |

**Futures Summary (F = Freighter; C = Colonist; T = TW Thruster)**

| Name                             | Src    | Requirements Summary                                                                                                                                                                             |
|----------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ Aerostat                       | C      | Have Promoted Bernal with Aerostat Dirtsides. Effect = Can homestead as free action as permanent ability, 14 VP.                                                                                 |
| ★ Antimatter                     | F      | At Promoted Bernal with S Dirtsides. Effects = double isotope refuel, 10 VP.                                                                                                                     |
| ★ Artificial Consciousness       | C      | 2 promoted Colonists at an Astrobiology Dirtsides. Effect = May free market any number of cards, 10 VP                                                                                           |
| ★ Beanstalk                      | F<br>C | No Beanstalk built, have 3+ Space Elevators built by any player. Effect = (Endgame) +3 VP for each Factory of any player connected to a Space Elevator.                                          |
| ★ Beehive Ark                    | F      | At Promoted Bernal anchored at a Synodic Comet. Effect = 7 VP.                                                                                                                                   |
| ★ Dyson Bubble                   | C      | Have both Sites of Mercury industrialized by any player. Effects = 5 VP per Factory owned on Mercury.                                                                                            |
| ★ Enzmann Starship               | T      | Ad astra exit with 2 Promoted Colonists + Mobile Factory, Effect = 12 VP.                                                                                                                        |
| ★ ET Life                        | C      | Have 2 or more Astrobiological Colonies. Effect (Endgame): +2 VP per Astrobiological Colony.                                                                                                     |
| ★ Exoplanet                      | F      | Claim Sedna. Effect = (Endgame) 12 VP.                                                                                                                                                           |
| ★ Footfall                       | C      | Decommission an operational thruster with net thrust 7+ on an Industrialized Synodic Comet of yours. Effects = 10 VP + Casus belli for War (1D2c). (Endgame) All tokens on the comet eliminated. |
| ★ Fusion Candle                  | T      | Have Triton Colony & Promoted Bernal with Neptune Aerostat Dirtsides. Effect = double all isotope refuel, 14 VP.                                                                                 |
| ★ Golden Apples                  | F      | Industrialize Kreutz Sungrazer. Effect = Ignore solar flares, 14 VP.                                                                                                                             |
| ★ Lithiated Ammonia Ice Starship | T      | Ad astra exit with 10 isotope fuel, Effect = 14 VP.                                                                                                                                              |
| ★ Mass Beam                      | T      | Have Promoted Bernal with Io or Triton Dirtsides. Effect: Your Powersat adds +2 thrust, 7 VP.                                                                                                    |
| ★ Mini-Black Hole                | T      | Have industrialized centaur, spend 10 isotope FTs. Effects = double all isotope refuel, 10 VP.                                                                                                   |
| ★ New Venus                      | C      | Decommission an operational thruster with net thrust 7+ on an Industrialized Synodic Comet of yours. Effects = 10 VP + (Endgame) All tokens on Venus & comet eliminated.                         |
| ★ Pan Sapiens                    | C      | Have 3 Factories connected to Space Elevators. Effects = Casus belli for War. (Endgame) +2 VP for each Glory chit.                                                                               |
| ★ Protium Fusion                 | T      | Have Promoted Bernal with H Dirtsides. Effects = double all isotope refuel, 10 VP.                                                                                                               |
| ★ Secession (New Attica)         | C      | Have 2 Promoted Human Colonists at a Promoted Anchored Bernal. Effects = Casus belli for War, 7 VP.                                                                                              |
| ★ Secession (Soldier)            | C      | 2 Promoted Human Colonists at an Anchored Bernal with Dirtsides 5+. Effects = Casus belli for War, 10 VP.                                                                                        |
| ★ SETI Future                    | C      | Industrialize 2 Jovian trojans, 1 each in the greek & trojan camps. Effect = As a free action perform 1 inspiration + 1 homestead, 10 VP.                                                        |
| ★ Spacefaring                    | T      | Have Bernal with 8+ dirtside hydration. Effect = Allowed 1 extra Colonist, 7 VP.                                                                                                                 |
| ★ Star Wisp                      | F      | Have Promoted freighter at either neutrino sunlens = 6VP or EM sunlens = 11 VP (Endgame)                                                                                                         |
| ★ Submariner                     | C      | Build 3 Submarines (Colonies or Factories). Effect = Doubles your dirtside hydration; not cumulative with other modifiers.                                                                       |
| ★ Supreme Cult                   | C      | Have Active Law in Authority. (Module 0) Effects = May lobby without removing the delegate used. All seniority disks migrate to authority. (Endgame) 10 VP.                                      |
| ★ Terraform                      | F      | At Promoted Bernal at a non-martian Atmospheric Dirtsides. Effect = 8 VP.                                                                                                                        |
| ★ TNO                            | C      | Industrialize 2 Sites in the Neptune Zone. Effect = Can homestead as a free action, 12 VP.                                                                                                       |
| ★ Uplift                         | C      | Robots not Emancipated. Human at a promoted Bernal & spend 20 Aqua. Effects = Every Robot becomes Emancipated, Casus belli for War, 12 VP                                                        |